Breno Cordeiro Bispo

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Summary

He received his Ph.D. in Electrical Engineering from the Federal University of Pernambuco (UFPE), Brazil in 2026 (CNPq-funded), with research focused on higher-order signal processing, computational neuroscience, and complex systems. He was a visiting researcher at the Korteweg–de Vries Institute (KdVI) and the Institute for Advanced Study (IAS) at the University of Amsterdam (UvA), The Netherlands, from November 2023 to August 2024 and from July to October 2025, supported by CAPES and the TSP-ARK project. During this period, he actively participated in several academic events, including workshops, seminars, and international conferences.

He received his M.Sc. in Electrical Engineering from UFPE in 2021 (CNPq-funded) and his B.Sc. in Electronic Engineering from UFPE in 2019. He was also a visiting undergraduate student at Hochschule Offenburg, Germany, under the Science Without Borders Program (CAPES-funded), where he studied Elektrotechnik/Informationstechnik from March 2015 to February 2016.

During his undergraduate studies, he served as a teaching assistant for courses in Electrical Circuits, Analog Electronics, and Digital Electronics. His professional experience includes biomedical sensors, embedded systems, Internet of Things (IoT), and FPGA-based hardware design.

His current research interests include (hyper)graph signal processing, topological signal processing, machine learning, neuroscience, and complex systems.

Academic Formation

2022 – 2026	**Ph.D. in Electrical Engineering** – UFPE, Brazil *Higher-Order Signal Processing and Its Applications in Brain Network Analysis* Advisors: Prof. J. B. Lima (UFPE) & Prof. F. A. N. Santos (UvA)	*granted by CNPq*
Nov. 2023 – Jul. 2024	**Visiting Researcher** – KdVI & IAS, University of Amsterdam (UvA) Supervisor: Prof. F. A. N. Santos (UvA)	*granted by CAPES*
Jul.–Oct. 2025	**Visiting Researcher** – Univ. of Amsterdam & Universitas Mercatorum (Rome) *Project: Topological Signal Processing: Applications in bRain and Knowledge networks (TSP-ARK)* Advisors: Prof. S. Sardellitti (Universitas Mercatorum) & Prof. F. A. N. Santos (UvA)	
2019 – 2021	**M.Sc. in Electrical Engineering** – UFPE, Brazil *Vital Sign Monitoring via NFC* Advisor: Prof. M. A. B. Rodrigues (UFPE)	*granted by CNPq*
2015 – 2016	**Exchange Undergraduate Student** – Hochschule Offenburg, Germany *CAPES – Science without Borders* Course: Elektrotechnik/Informationstechnik	
2012 – 2019	**B.Sc. in Electronic Engineering** – UFPE, Brazil *Access Control System via RFID/NFC* Advisor: Prof. M. A. B. Rodrigues (UFPE)	

PhD Description

The doctoral thesis, entitled *Higher-Order Signal Processing and Its Applications in Brain Network Analysis* (UFPE, 2026; advisors: Prof. Juliano B. Lima and Prof. Fernando A. N. Santos), addresses a central limitation of graph-based models. While graphs provide a powerful language for modeling dyadic relations, many real-world complex systems—and, in particular, the human brain—exhibit *polyadic* dependencies involving groups of interacting elements. Such many-body interactions cannot be faithfully represented by pairwise graphs alone. Motivated by this gap, the thesis develops a progressive and unified higher-order signal processing framework for signals defined over hypergraphs, simplicial complexes, and cell multicomplexes, with a particular focus on interpretable applications to brain functional connectivity.

Research trajectory and contributions:

- **Hypergraph Signal Processing (HGSP) for brain networks:** The first axis of the thesis extends spectral tools from GSP—including Fourier analysis, filtering, and community detection—to hypergraph domains, where a single hyperedge can encode interactions among multiple brain regions. This framework combines hypergraph representations with multivariate information-theoretic measures to uncover higher-order functional communities in resting-state fMRI data. The results show that hypergraph-based models reveal integrative community structures and cross-network interaction patterns that are systematically missed or only partially captured by conventional graph-based methods. This line of work established the first methodological layer of the thesis and resulted in one journal article in *Computers in Biology and Medicine* and one conference paper at EUSIPCO 2024 (Lyon).
- **Topological Signal Processing (TSP) over cell multicomplexes:** Developed in collaboration with Prof. Stefania Sardellitti (Universitas Mercatorum, Rome) during a visiting research period funded by the TSP-ARK project (2025), this line of work extends signal processing tools to cell multicomplexes. Novel cross-Laplacian operators were introduced to process signals *across* cells of different dimensions, such as nodes, edges, and higher-order cells. This provided a unified algebraic machinery for modeling interactions between layers, modalities, and topological dimensions. The results led to a journal submission to *IEEE Transactions on Signal Processing* (arXiv:2510.09139) and one paper at EUSIPCO 2025 (Palermo).

- **Multimodal higher-order brain network analysis:** The culminating contribution establishes a topological signal processing (TSP) framework for multimodal higher-order brain network analysis. By jointly integrating structural connectivity, functional edge dynamics, and subject-specific higher-order topology, this work moves beyond conventional pairwise graph representations and enables brain signals to be analyzed over cell complex domains. We introduce TSP and discrete vector calculus operators into brain network analysis, providing neuroscientific interpretations for gradient, divergence, curl, and harmonic components in the brain. Within this framework, divergence reveals source–sink organization across functional systems, curl captures cyclic coordination patterns. This contribution is presented in the preprint *Multimodal Higher-Order Brain Networks: A Topological Signal Processing Perspective* (arXiv:2603.29903), submitted to the *IEEE Journal of Biomedical and Health Informatics*, with preliminary results published at EUSIPCO 2025 (Palermo).

Overall, the thesis follows a coherent scientific trajectory: from pairwise graph-based signal processing, to hypergraph representations of polyadic functional interactions, and finally to topological signal processing frameworks capable of extracting interpretable discrete vector calculus features from multimodal brain networks. The work was carried out at UFPE, with an international period at the Korteweg–de Vries Institute (KdVI) and the Institute for Advanced Study (IAS) at the University of Amsterdam (Nov. 2023–Jul. 2024, CAPES-funded), followed by a visiting research stay at UvA and Universitas Mercatorum (Jul.–Oct. 2025, TSP-ARK project).

Research Interests

Topological Signal Processing • (Hyper)graph Signal Processing • Machine Learning • Neuroscience • Complex Systems • Internet of Things • Embedded Systems

Complete List of Publications

The three most significant publications are highlighted below.

Journal Articles Under Review

★ Most Significant Publication #1

BISPO, B.C.; SARDELLITTI, S.; SANTOS, F.A.N.; LIMA, J.B. (2026) "Multimodal Higher-Order Brain Networks: A Topological Signal Processing Perspective." submitted to *IEEE Journal of Biomedical and Health Informatics*. `https://arxiv.org/abs/2603.29903`

It introduces topological signal processing and discrete vector calculus operators into multimodal brain network analysis, translating gradient, divergence and curl components into interpretable signatures of brain function. The framework jointly integrates structural connectivity, functional edge dynamics, and subject-specific higher-order topology to characterize source–sink organization, cyclic coordination, and non-gradient functional patterns.

★ Most Significant Publication #2

SARDELLITTI, S.; **BISPO, B.C.**; SANTOS, F.A.N.; LIMA, J.B. (2025) "Topological Signal Processing Over Cell MultiComplexes Via Cross-Laplacian Operators." submitted to *IEEE Transactions on Signal Processing*. `https://arxiv.org/abs/2510.09139`

This work introduces Cell MultiComplexes as topological spaces for representing higher-order intra- and inter-layer interactions in multilayer networks. By defining cross-Laplacian operators, the framework enables topological signal processing over cross-layer domains, including spectral representations, Hodge-based decompositions, filtering of noisy cross-edge flows, and topology learning. The results show that cross-Laplacian bases provide sparse and localized representations of multilayer signals, with applications to the inference of inter-module higher-order structures in brain networks.

- LIMA, J.B.; SILVA NETO, E.F.; **BISPO, B.C.**; MORGADO, P. (2026) "Schizophrenia Detection from Electroencephalogram Signals Using Graph Permutation Entropies." submitted to *NeuroImage*.
- SILVA NETO, E.F.; **BISPO, B.C.**; LIMA, J.B. (2026) "Tensor-Based Hypergraph Learning for Schizophrenia and ADHD Classification." submitted to *IEEE Transactions on Image Processing*.

Journal Articles

★ Most Significant Publication #3

BISPO, B.C.; DE OLIVEIRA NETO, J.R.; LIMA, J.B.; SANTOS, F.A.N. (2026) "From Pairwise to Higher-Order Clustering: A (Hyper-)Graph Signal Processing Approach on Brain Functional Connectivity Analysis." *Computers in Biology and Medicine*. `https://doi.org/10.1016/j.compbiomed.2025.111409`

It introduces a hypergraph signal processing framework for higher-order brain community detection, integrating spectral clustering with multivariate information-theoretic measures to move beyond conventional pairwise functional connectivity models. Through empirical, surrogate, and synthetic fMRI analyses, the study shows that hypergraph representations uncover integrative functional communities and cross-network interactions that remain partially hidden in graph-based approaches.

- LIMA, J.B.; **BISPO, B.C.**; NETO, E.F.S.; MORGADO, P. (2026) "Novel Measures of Pairwise Time Series Interdependence Using Graph Permutation Entropies: An Application to Brain Functional Connectivity." *Computers in Biology and Medicine*. `https://doi.org/10.1016/j.compbiomed.2026.111682`

- **BISPO, B.C.**; DE OLIVEIRA NETO, J.R.; LIMA, J.B. (2023) "Hardware Architectures for Computing Eigendecomposition-Based Discrete Fractional Fourier Transforms with Reduced Arithmetic Complexity." *Circuits, Systems, and Signal Processing*, v.42. https://doi.org/10.1007/s00034-023-02493-1

Conference Papers and Book Chapters

- SILVA NETO, E.F.; BONAFINI, B.L.; **BISPO, B.C.**; LIMA, J.B. (2026) "Comparing MedSigLIP and Structured Connectivity Models for ADHD and Schizophrenia Classification." In: *Simpósio Brasileiro de Computação Aplicada à Saúde (SBCAS)*, 26., Ouro Preto/MG, pp. 1391–1396. https://doi.org/10.5753/sbcas.2026.21443
- **BISPO, B.C.**, SARDELLITTI, S., SANTOS, F.A.N., LIMA, J.B. (2025). "Learning Higher-Order Interactions in Brain Networks via Topological Signal Processing." *33rd European Signal Processing Conference (EUSIPCO)*, Palermo, Italy, pp. 2652–2656. https://doi.org/10.23919/EUSIPCO63237.2025.11226759
- SARDELLITTI, S.; **BISPO, B.C.**; SANTOS, F.A.N.; LIMA, J.B. (2025) "Cross-Laplacians Based Topological Signal Processing over Cell Multicomplexes." *33rd European Signal Processing Conference (EUSIPCO)*, Palermo, Italy, pp. 2647–2651. https://doi.org/10.23919/EUSIPCO63237.2025.11226604
- **BISPO, B.C.**; SANTOS, F.A.N.; DE OLIVEIRA NETO, J.R.; LIMA, J.B. (2024) "Emergence of Higher-Order Functional Brain Connectivity with Hypergraph Signal Processing." *32nd European Signal Processing Conference (EUSIPCO)*, Lyon, France, pp. 1332–1336. https://doi.org/10.23919/EUSIPCO63174.2024.10715376
- **BISPO, B.C.**; CAVALCANTE, E.L.; RODRIGUES, M.A.B. (2024) "Access Control System Integrated with RFID and NFC-Enabled Smartphone Technologies." In: *IFMBE Proceedings*, 9th ed., Springer Nature Switzerland, v. 100, pp. 657–667. https://doi.org/10.1007/978-3-031-49407-9_65
- **BISPO, B.C.**; CUNHA, N.A.; SILVA, M.C.B.C.; RODRIGUES, M.A.B. (2024) "Batteryless NFC Device for Heart Rate and SpO2 Acquisition." In: *IFMBE Proceedings*, 9th ed., Springer Nature Switzerland, v. 100, pp. 465–475. https://doi.org/10.1007/978-3-031-49407-9_47
- SILVA, A.T.D.; **BISPO, B.C.**; ESTEVES, G.R.P.; SILVA, M.C.B.C.; RODRIGUES, M.A.B. (2022) "A Serious Game that Can Aid Physiotherapy Treatment in Children Using Dennis Brown Orthotics." In: *XXVII CBEB – IFMBE Proceedings*, Springer, v. 83, pp. 445–450. https://doi.org/10.1007/978-3-030-70601-2_69
- **BISPO, B.C.**; CAVALCANTE, E.L.; ESTEVES, G.R.P.; SILVA, M.C.B.C.; ALVES, G.J.; RODRIGUES, M.A.B. (2022) "Access Control in Hospitals with RFID and BLE Technologies." In: *XXVII CBEB – IFMBE Proceedings*, Springer, v. 83, pp. 917–924. https://doi.org/10.1007/978-3-030-70601-2_137
- SILVA, A.T.D.; **BISPO, B.C.**; CONCEIÇÃO, A.M.; SILVA, M.C.B.C.; ESTEVES, G.R.P.; RODRIGUES, M.A.B. (2022) "Adapted Children's Serious Game Using Dennis Brown Orthotics During the Preparatory Phase for the Gait." In: *XXVII CBEB – IFMBE Proceedings*, Springer, v. 83, pp. 439–443. https://doi.org/10.1007/978-3-030-70601-2_68
- CUNHA, N.A.; **BISPO, B.C.**; CAVALCANTE, E.L.; INOCENCIO, A.V.M.; ALVES, G.J.; RODRIGUES, M.A.B. (2022) "Application of MQTT Network for Communication in Healthcare Establishment." In: *XXVII CBEB – IFMBE Proceedings*, Springer, v. 83, pp. 1465–1470. https://doi.org/10.1007/978-3-030-70601-2_216
- SILVA, M.C.B.C.; **BISPO, B.C.**; SILVA, C.M.; CUNHA, N.A.; SANTOS, E.A.B.; RODRIGUES, M.A.B. (2022) "Assisted Navigation System for the Visually Impaired." In: *XXVII CBEB – IFMBE Proceedings*, Springer, v. 83, pp. 1439–1444. https://doi.org/10.1007/978-3-030-70601-2_212
- CUNHA, N.A.; **BISPO, B.C.**; FERREIRA, K.R.C.; ALVES, G.J.; ESTEVES, G.R.P.; SANTOS, E.A.B.; RODRIGUES, M.A.B. (2022) "Communication in Hospital Environment Using Power Line Communications." In: *XXVII CBEB – IFMBE Proceedings*, Springer, v. 83, pp. 1451–1455. https://doi.org/10.1007/978-3-030-70601-2_214
- SILVA, A.T.D.; SILVEIRA, J.V.B.D.; ESTEVES, G.R.P.; **BISPO, B.C.**; CUNHA, N.A.; KUNKEL, M.E.; RODRIGUES, M.A.B. (2022) "Development and Customization of a Dennis Brown Orthosis Prototype Produced from Anthropometric Measurements by Additive Manufacturing." In: *XXVII CBEB – IFMBE Proceedings*, Springer, v. 83, pp. 465–470. https://doi.org/10.1007/978-3-030-70601-2_72
- CHAVES FILHO, A.C.; INOCENCIO, A.V.M.; FERREIRA, K.R.C.; CAVALCANTE, E.L.; **BISPO, B.C.**; RODRIGUES, C.M.B.; LESSA, P.S.; RODRIGUES, M.A.B. (2022) "Geriatric Physiotherapy: A Telerehabilitation System for Identifying Therapeutic Exercises." In: *XXVII CBEB – IFMBE Proceedings*, Springer, v. 83, pp. 969–974. https://doi.org/10.1007/978-3-030-70601-2_144
- SILVA, C.M.; **BISPO, B.C.**; ESTEVES, G.R.P.; CAVALCANTE, E.L.; OLIVEIRA, A.L.B.; SILVA, M.C.B.C.; CUNHA, N.A.; RODRIGUES, M.A.B. (2022) "Transducer for the Strengthening of the Pelvic Floor Through Electromyographic Biofeedback." In: *XXVII CBEB – IFMBE Proceedings*, Springer, v. 83, pp. 935–940. https://doi.org/10.1007/978-3-030-70601-2_139
- **BISPO, B.C.**; CUNHA, N.A.; CAVALCANTE, E.L.; ESTEVES, G.R.P.; FERREIRA, K.R.C.; CHAVES FILHO, A.C.; RODRIGUES, M.A.B. (2022) "Wearable System for Postural Adaptation." In: *XXVII CBEB – IFMBE Proceedings*, Springer, v. 83, pp. 387–393. https://doi.org/10.1007/978-3-030-70601-2_60

Research Projects

2024 – present **Graph Signal Processing: Theoretical Contributions and Application Scenarios**
PI: Prof. Juliano B. Lima (UFPE)
Develops tools that generalise classical signal processing concepts to signals defined over graph-structured domains (sensor networks, brain activity, digital images), in a cooperative network involving Brazilian and international universities.

2024 – 2026 **TSP-ARK: Topological Signal Processing – Applications in bRain and Knowledge Networks**
PIs: Prof. Stefania Sardellitti (Universitas Mercatorum) & Prof. Fernando A. N. Santos (UvA)
Develops novel topological signal processing methods with applications to brain network analysis.

2022 – 2024 **Contributions to Signal Processing with Applications to IoT and Digital Image Representation**
PI: Prof. Juliano B. Lima (UFPE)
Advances graph signal processing theory targeting applications in IoT sensor networks and digital image representation.

2020 – 2021 **Monitoring of Therapeutic Exercises and Haemodynamic Parameters for Post-COVID-19 Patients**
PI: Prof. Marco A. B. Rodrigues (UFPE)
Developed a wearable device for non-invasive monitoring of vital signs and therapeutic exercises via accelerometry and imaging, targeting remote rehabilitation of post-hospitalisation COVID-19 patients.

Talks, Presentations & Workshops

- SARDELLITTI, S.; **BISPO, B.C.**; SANTOS, F.A.N.; LIMA, J.B. (2026) "Cross-Laplacians Based Topological Signal Processing over Cell MultiComplexes." *Graph Signal Processing Workshop 2026*, Madrid, Spain.
- "Network Inference." (2026) *CIMPA Research School – The Mathematics of Complex Systems*, Recife, Brazil.
- "Application of Topological Signal Processing to Brain Networks." (2025) *Cutting-Edge Perspectives on Brain Networks Workshop*, Universitas Mercatorum, Rome, Italy.
- "An Overview on Hypergraph Signal Processing and Its Application in Neuroscience." (2024) *KdVI DM Seminar*, University of Amsterdam.
- **BISPO, B.C.**; SANTOS, F.A.N.; OLIVEIRA NETO, J.R.; LIMA, J.B. (2024) "Emergence of Higher-Order Functional Brain Connectivity with Hypergraph Signal Processing." *Graph Signal Processing Workshop 2024*, Delft, Netherlands.

Professional Experience

Mar.–Dec. 2018 **Engineering Intern** – EngeBIO NE, Recife · RFID/NFC-based access control system development.
Feb.–Jun. 2020 **Teaching Assistant** – UFPE · Digital Electronics 1A (internship).
2014 – 2017 **Teaching Monitor** – UFPE · Electric Circuits (2014); Digital Electronics (2016–17); Analog Electronics (2017).

Technical Skills

Programming	C/C++, Python, MATLAB
Hardware Description	VHDL, Verilog
Dev. Environments	Arduino IDE, Quartus, KiCAD, LTSpice
Microcontrollers	ESP8266, ESP32, ATMEGAs
FPGA Platforms	Altera DE2-115, DE1-SoC
IoT Protocols	MQTT, LoRa, Bluetooth/BLE, WiFi, Ethernet, nRF24L01
OS / General	Windows, Linux

Languages

Portuguese – Native **English** – Fluent (TOEFL ITP B2) **German** – Intermediate (TELC B1)

Awards

2017 **SBrT Digital Communications Challenge for Military Applications** – Brazilian Telecommunications Society (SBrT).